Date:- 24/9/25 Task - 9.1

a. Student marks validation.

AIM:

To write a python program that accepts student marks, raises exceptions if the entered marks are invalid (less than 0 or greater than or equal to 100), and handles them with appropriate error messages.

ALGORITHM:

1. start
2. Accept marks from the user as input.
3. Convert the input into an integer.
4. If marks< 0 or marks >= 100,
   → Raise an exception.
5. Otherwise, display the marks.
6. Catch exceptions and display appropriate error messages.
7. stop.

Program:

```
# student marks validation
try:
    marks = int( input( "Enter student marks(0-99): "))
    if marks<0 or marks >=100:
        rasie Valueerror("marks should be between 0 and 99 only!")
    print("marks entered successfully :", marks)
except Valueerror as e:
    print("Error:", e)
```

Sample output:

Enter student marks (0-99) : 85
Marks entered successfully : 85

Enter student marks (0-99) : 120
Error : marks should be between 0 and 99 only!

Enter student marks (0-99) : -5
Error : marks should be between 0 and 99 only!

RESULT: Tho program successfully valians student marks and displays an error message for invalid input.

24/9/25 Task 9.2

(b) Division calculator with Exception Handling

AIM:

To write a python program that accepts two numbers from the user, perform division, and handles exceptions such as division by zero and invalid input.

ALGORITHM

1- start

2. Accept two numbers (numerator and denominator) from the user

3. Convert both inputs to integers.

4- perform division → result = numerator/denominator.

5. If denominator is zero → handle zero DivisionError.

6. If input is invalid (not a number) → handle valueError.

7. Display result or appropriate error message

8. Stop.

program:

python

```
# Division calculator with Exception Handling
try:
    num1 = int(input("Enter numerator:"))
    num2 = int(input("Enter denominator:"))

    result = num1 / num2
    print("Result of division:", result)
Except zeroDivisionError:
Print("Error: Division by zero is not allowed!")
except valueError:
    print("Error: please enter valid integers!")
```

Samplo output:

Enter numerator : 10

Enter denominator : 2

Result of division : 5.0

Enter numerator : 10

Enter denominator : 0

Error : Division by zero is not allowed!

Enter numerator : 15

Enter denominator : abc

Error: please enter valid integers!

VEL TECH-CSE

EX NO.	9
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	15
SIGN WITH DATE	15/10/25

RESULT:

The program correctly perform division and handles both invalid and division by zero errors

15/10/25